

Eccentric Counts, Connectivity and Chordality

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Abstract

Let e_j denote the number of vertices of eccentricity j in a graph. We provide sharp lower bounds on e_j in graphs of given connectivity and in chordal graphs of given connectivity.

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1 Introduction

The eccentricity of a vertex v is the maximum distance of a vertex from v , and the eccentricity sequence of a graph is the sequence of the eccentricities of its vertices (sorted in nondecreasing order). In 1975, Lesniak [7] characterized the sequences of integers that are the *eccentric sequences* of trees. However, a characterization of eccentric sequences for graphs in general has not been found. Indeed, partial results suggest that the problem is complex (see, for example, [5, 6]). We consider here a related question, namely, bounds on the number of vertices with a given eccentricity. In this regard, Mubayi and West [8] gave sharp lower bounds and almost sharp upper bounds on the number of vertices of given eccentricity in terms of order and diameter.

We will denote the eccentricity of vertex v by $\text{ecc}(v)$. We use E_j for the set of vertices of eccentricity j , and we let $e_j = |E_j|$ (the j^{th} eccentricity count). Further, the radius r is the minimum eccentricity and the diameter d is the maximum eccentricity.

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It is easy to show (see for example [7]) that in a connected graph, $e_j \geq 2$ for all j with $r < j \leq d$. Indeed, since E_j is a cut-set when $r < j < d$, it follows that

For $r < j < d$, $e_j \geq k$ in a k -connected graph.

In this paper we consider improvements on this result for some graph classes. In section 2 we provide a sharp lower bound on e_j in k -connected graphs, and in section 3 we provide a sharp lower bound on e_j in k -connected chordal graphs. Finally, in section 4 we consider the question for a generalization of chordal graphs.

2 Connectivity

We begin with a lower bound for the eccentricity count e_j in general graphs based on the connectivity. The results shows that in a k -connected graph, the eccentricity count e_j can be less than $2k$ only if j is not too close to $2r$. As a corollary we obtain a lower bound on e_j in terms of connectivity.

Theorem 1. *Let G be a k -connected graph with radius r and diameter d , and let j be such that $r < j < d$. If $e_j = s$ and $s < 2k$, then*

$$j \leq \frac{s}{s+1} 2r + \frac{s-1}{2(s+1)},$$

and this is sharp.

Proof. Let $e_j = s$ and let $E_j = \{a_1, \dots, a_s\}$. Let F be the set of vertices of G with eccentricity more than j . Since $j < d$, this set is nonempty. Every vertex of F is within distance r of the center, while every vertex of E_j is at least distance $j - r$ from the center. Thus

every vertex of F is within distance $r - (j - r) = 2r - j$ of E_j .

For $1 \leq i \leq s$, define F_i as the set of vertices in F that are within distance $2r - j$ of a_i via a path that does not contain any other vertex of E_j . Note that $\bigcup_i F_i = F$.

Now, define an auxiliary graph H as follows. Create one vertex w_i for each set F_i that is nonempty. Then add an edge from w_i to $w_{i'}$, if there is a vertex in $F_i \cap F_{i'}$ or there is in G an edge from F_i to $F_{i'}$.

We claim that H is connected. For, let H' be any component of H . Note that by the way we defined F_i , each induced subgraph $\langle F_i \cup \{a_i\} \rangle$ of G is connected. Further, if $x \in F_i$ is adjacent to some $a_{i'}$, then $x \in F_{i'}$, so that $w_{i'}$ is a vertex in H and there is an edge in H from w_i to $w_{i'}$. It follows that $\{a_i : w_i \in H'\}$ is a cut-set in G . By the connectivity of G , it follows that this set has at least k vertices, and so H' has at least k vertices. Since H has less than $2k$ vertices, it follows that H is connected.

Now, let R be the radius of H and let w_{i^*} be a central vertex in H . Let x be any vertex of F_{i^*} adjacent to a_{i^*} . We can bound the eccentricity in G of x as follows.

Let y be an eccentric vertex for x (that is, y is farthest from x). We know that $y \in F$, since all other vertices have eccentricity at most j . Say $y \in F_{i'}$. Then a path in G from x to y can be constructed within the induced subgraph $\langle F \cup E_j \rangle$ by mimicking the path in H from w_{i^*} to $w_{i'}$. Since each induced subgraph $\langle F_i \cup \{a_i\} \rangle$ has diameter at most $2(2r - j)$, vertex x has eccentricity at most $1 + (2r - j)$ within the induced subgraph $\langle F_{i^*} \cup \{a_{i^*}\} \rangle$, and H has radius R , it follows that

$$\text{ecc}(x) \leq 1 + (2r - j) + R(2(2r - j) + 1).$$

But, $\text{ecc}(x) \geq j + 1$, because $x \in F$. It follows that

$$j \leq \frac{r(4R + 2) + R}{2R + 2}.$$

Note that this upper bound is increasing in R . If we set $R = (s - 1)/2$ into the upper bound and simplify, then we obtain the formula in the statement of the theorem. Thus, if $R \leq (s - 1)/2$, we are done.

So assume that $R > (s - 1)/2$. This means that $R = s/2$. It follows that s is even and H is a path on s vertices. Therefore, there are two central vertices in H , say w_{i^*} and $w_{i^{**}}$. Since these two vertices are adjacent in H , there is a vertex x' in F_{i^*} that is either in $F_{i^{**}}$ or adjacent to a vertex of $F_{i^{**}}$.

By a similar argument to the above, it follows that

$$\text{ecc}(x') \leq (s/2)(2(2r - j) + 1).$$

And since $\text{ecc}(x') \geq j + 1$, we get that

$$j \leq \frac{2sr + (s/2 - 1)}{s + 1},$$

as required. (The bound for $R = (s - 1)/2$ appears larger, but as s is even, that bound is not an integer and is not exactly attainable.) This proves the theorem. $\square$

Corollary 2. *Let G be a k -connected graph with radius r and diameter d , and let j be such that $r < j < d$. Then*

$$e_j \geq \left\lceil \frac{2j + 1}{4r - 2j + 1} \right\rceil_k^{2k},$$

where by $[A]_k^{2k}$ we mean $\min\{\max\{A, k\}, 2k\}$.

Theorem 1 is best possible. First we note that it is easy to construct k -connected graphs where $e_j = 2k$. For example, take the power P_n^k of the path for n sufficiently large.

Further, the upper bound on j is best possible in general. Here is an example of an extremal graph. Fix positive integers r, j, s, k such that $s \geq 2$, $r < j < 2r - 1$, and $j \leq (2sr + (s - 1)/2)/(s + 1)$. Construct graph $G(r, j, s, k)$ as follows.

Start with a rooted tree with root c where c has s neighbors, $a_1, \dots, a_s$, each of which has two leaf-neighbors b_i and b'_i . Then subdivide each edge ca_i to have length $j - r$ and subdivide every other edge to have length $2r - j$. Add an edge from b'_i to b_{i+1} for each $1 \leq i \leq s - 1$. Add a path of length 3 joining the two child neighbors of a_i for each $1 \leq i \leq s$. Finally, expand every vertex v except the a_i into a clique C_v of order k in the natural way: a vertex in C_v is adjacent to a vertex in $C_{v'}$ if and only if v and v' were adjacent, and a vertex in C_v is adjacent to a_i if and only if v and a_i were adjacent.

Note that vertices in C_c have eccentricity r , and so a_i has eccentricity at most j . Let $T = \{b_1, b'_s\}$, and for any vertex v , define $X(v) = \max\{d(v, b_1), d(v, b'_s)\}$. Consider any of the a_i . It follows that the shortest path in $G - c$ from a_i to the

farther vertex of T has length at least $s(2r - j) + (s - 1)/2$. By the upper bound on j , it follows that $X(a_i) \geq j$, and so $\text{ecc}(a_i) = j$.

Every vertex on the subdivided edge ca_i has eccentricity less than j , since it is at distance less than $j - r$ from C_c . Further, let w be any descendant of the $\{a_i\}$. Any path from w to a vertex of T that goes via C_c has length more than j . Further, $X(w) > X(a_i)$ for some a_i , since the shortest C_c -avoiding path from w to the farther vertex of T meets the shortest C_c -avoiding path from some a_i to that vertex of T at a point closer to a_i than to w . It follows that $\text{ecc}(w) > j$.

Hence $E_j = \{a_1, \dots, a_s\}$. At the same time, $G(r, j, s, k)$ has connectivity $\min\{s, k\}$. In Figure 1 we depict $G(10, 17, 5, k)$.

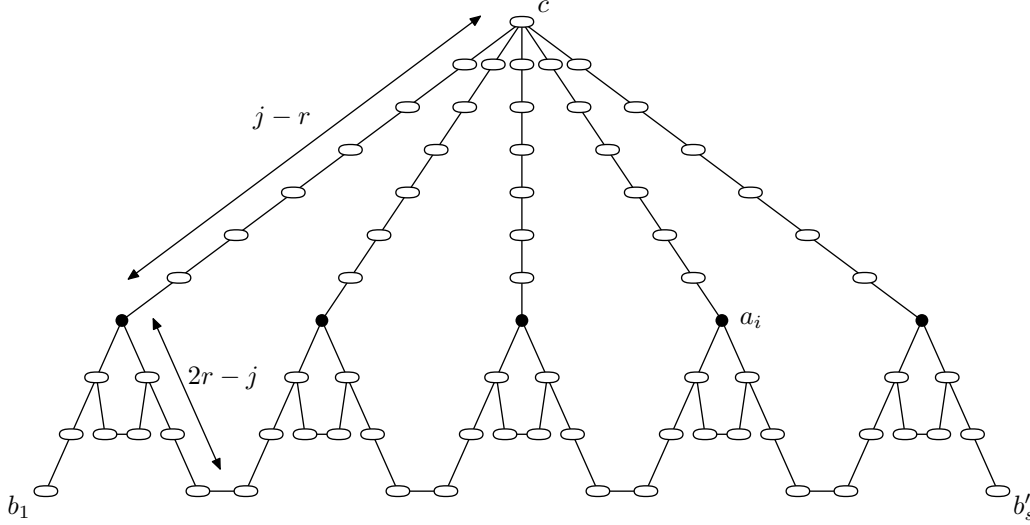


Figure 1: The graph $G(10, 17, 5, k)$ with $e_{(5r+1)/3} = 5$.

In passing, we note that there is no equivalent result for edge connectivity. That is, one can readily construct graphs with arbitrary edge connectivity and still have $e_j = 2$ for specific j . For example, consider the graph obtained from a path of suitable length by expanding every vertex except the two vertices of eccentricity j into a large clique.

3 Chordal Graphs

Recall that a graph is *chordal* if every cycle of length at least 4 has a chord. We will need the following well-known facts about chordal graphs (a separator separates some pair of nonadjacent vertices) (see, for example, [1]):

Proposition 3. *If G is chordal graph, then*

- (a) *G has a simplicial vertex;*
- (b) *every minimal separator is a clique.*

We will also need the following simple result, which is probably known.

Lemma 4. *Let G be a k -connected chordal graph, and S a cut-set of G . Then in each component of $G - S$ there is a vertex v such that $N(v) \cap S$ contains a k -clique.*

Proof. Recall the well-known fact that for any graph G , if vertex w is simplicial and G is not complete, then the removal of w cannot decrease the connectivity.

Consider any simplicial vertex w . If $w \in S$, then w has neighbors in at most one component of $G - S$. So we can induct on $G - w$; the vertex v so found has the desired property in G also.

So assume $w \notin S$. If w is not an isolated vertex in $G - S$, then again we can induct on $G - w$. But if w is an isolated vertex in $G - S$, then its at least k neighbors are in S and form a clique. $\square$

Theorem 5. *Let G be a k -connected chordal graph with radius r and diameter d , and let j be such that $r < j < d$. Then*

$$e_j \geq 2k,$$

and this is sharp.

Proof. Say $E_j = \{a_1, \dots, a_s\}$ and suppose $s < 2k$. Let F be the set of vertices of G with eccentricity more than j . By the above lemma, there exists a vertex x in F that has k neighbors in E_j .

Let c be a central vertex and let y be an eccentric vertex for x . Then $y \in F$. Since the set E_j separates vertex y from c , there is a minimal subset C_y of E_j separating y from c . By chordality, C_y is a clique. Further, y is within distance r of c , while every vertex of E_j is at least distance $j - r$ from c , and so y is within distance $r - (j - r) = 2r - j$ of C_y .

Now, since $s < 2k$, it follows that $N(x)$ intersects the clique C_y . Thus $d(x, y) \leq 2 + (2r - j)$, and so $\text{ecc}(x) \leq 2 + 2r - j$. But $\text{ecc}(x) \geq j + 1$. It follows that $j \leq r + 1/2$, a contradiction. $\square$

It is easy to construct k -connected chordal graphs where $e_j = 2k$. For example, take the power P_n^k of the path for n sufficiently large, as before.

4 Chordal Generalizations

The key ingredient in the proof of Theorem 5 is that a minimal separator is a clique. One can generalize this theorem to graphs where there is some upper bound on the diameter of a minimal separator (diameter measured in the original graph). One such family is the k -chordal graphs. For $k \geq 2$, a graph is k -chordal if there is no induced cycle with length bigger than k (so that chordal graphs are 3-chordal and trees are 2-chordal).

Chepoi et al. [4] observed that:

Proposition 6. *If graph G is ℓ -chordal, then the diameter of any minimal separator is at most $\ell/2$.*

Using this, we can establish the following result:

Theorem 7. *If G is a k -connected ℓ -chordal graph, then*

- (a) $d \geq 2r - \ell - 1$.
- (b) $e_j \geq 2k$ for $d > j > r + (\ell + 1)/2$.

Proof. The result is immediate if G is complete; so assume otherwise. Let x and y be diametrical vertices. Let X be the vertices within distance $\lfloor d/2 \rfloor - 1$ of x , and Y the vertices within distance $\lceil d/2 \rceil - 1$ of y . Then $V - (X \cup Y)$ is a separator, and it contains a minimal (x, y) -separator S .

Every vertex v in the graph is within distance d of both x and y . Since the shortest path from v to at least one of x or y must go through S , it follows that vertex v is within distance $d - \lfloor d/2 \rfloor$ of S . Thus every vertex in S has eccentricity at most $\lceil d/2 \rceil + \ell/2$. This yields part (a).

Further, consider the $|S| \geq k$ internally disjoint paths between x and y . These paths go through S , where their vertices have eccentricity at most $\lceil d/2 \rceil + \ell/2$. So, assume $j > r + (\ell + 1)/2$. It follows that each such path must contain two vertices of eccentricity j , so that $e_j \geq 2k$. $\square$

The above bound is probably not sharp in terms of ℓ , since for example, it is known that $d \geq 2r - 2$ in chordal graphs (see [3, 2, 9]). Nevertheless, in part (b) some lower bound on j is necessary to force $e_j \geq 2k$, as the results in section 2 showed.

5 Open Questions

The ultimate goal is to characterize eccentric sequences of graphs. In order to make progress, perhaps one can obtain improved bounds for e_j for other graphs classes, or even characterize the eccentric sequences of some graphs classes.

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